Sudha Rani Maddala

AI-ML Enthusiast

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Bristol, UK PSW VISA - JAN 2027

Professional Summary

Innovative Data Scientist with a strong foundation in AI, ML model optimization, and computational models. Experienced in model compression, quantization, and optimization techniques, including post-training quantization and quantization-aware training for large language models (LLMs). Adept at leveraging PyTorch, TensorFlow, and Python to deliver impactful solutions in model performance enhancement. Eager to contribute to cutting-edge research and the optimization of ML models in the cloud infrastructure domain.

Technical Skills

Programming Languages: Python, SQL
Tools & Technologies: TensorFlow, Keras, Scikit-learn, PyTorch, Tableau, Excel
Cloud: Amazon Web Services (AWS), Google Cloud Platform
Development Tools: Git, LaTeX, Markdown
Key Skills: Data Analysis, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Visualization, Statistical Analysis
Soft Skills: Problem-solving, Communication, Team Collaboration, Adaptability
Languages: English, Hindi, Telugu

Experience

Software Developer - Freelancing — Aug 2026 – Present

4 Real Consulting Limited · Remote

- Developed a full-stack social-impact analytics dashboard using React/TypeScript, Python/FastAPI, and PostgreSQL, centralising organisational reporting data across attendance, service activity, strategic missions, and financial metrics.
- Built and structured backend data services and REST APIs to ingest, transform, and expose reporting metrics for interactive dashboard consumption.
- Analysed organisational reporting data and developed KPIs, comparative analytics, activity heatmaps, quarterly reporting, and revenue-split analysis to support stakeholder decision-making.
- Designed the reporting architecture with clear data provenance, separating source organisational data from business-case estimates to improve transparency and reliability of analytical outputs.
- Developed accessibility-focused software features including adjustable text sizing, high-contrast support, keyboard-friendly controls, and a persistent accessibility panel.
- Structured the platform to support future AI-assisted insights and automated reporting, while keeping deterministic metric calculations separate from generated interpretation.

Applied AI Research – Vision Assist — Nov 2025 – Jul 2026

Bristol, UK

- Conducted accessibility and user research to understand mobility challenges and navigation needs of blind and visually impaired users, informing the design of an AI-assisted mobility solution.
- Analysed mobility workflows and accessibility requirements to identify opportunities for computer vision, real-time hazard detection, and voice-based guidance.
- Developed a real-time AI prototype using Python, YOLO, and OpenCV, integrating low-latency voice guidance using smartphone-based hardware.
- Explored lightweight inference approaches and iteratively refined the system around usability, accessibility, and cognitive-load considerations.
- Used user feedback and accessibility research to evaluate practical usability and guide subsequent development.

AI-ML Intern — Sep 2025 – Oct 2025

Labmentix · Remote *Internship*

- Worked on real-world projects in Data Science and AI/ML, focusing on end-to-end problem-solving and analytics.
- Built predictive models and analyzed crime data trends for the **FBI Time Series Project**.

- Derived user and fitness insights from activity data in the **Strava Fitness App Analytics** project.
- Optimized store performance using data-driven insights for **Integrated Retail Analytics**.
- Applied ML for financial data analysis and customer insights in the **Paisabazar Project**.
- Strengthened expertise in Python, Machine Learning, Data Analysis, and Business Analytics with hands-on datasets.

Artificial Intelligence Intern May 2025 – Jul 2025
Personifwy · Remote *Internship*

- Developed and tested machine learning models to extract data-driven insights.
- Applied AI concepts to improve automation and efficiency in real-world projects.
- Collaborated with mentors to design practical AI use cases.
- Enhanced skills in Python, data science workflows, and problem-solving.

Project Development Intern May 2024 – Jul 2024
Streams Learning Hub · Bristol, UK *Internship*

- Created an educational portfolio for flexible learning environments tailored to neurodivergent students aged 12+.
- Researched AI applications for personalized education, including recommendation systems and NLP feedback analysis.
- Conducted market research and literature reviews on emerging educational technologies.
- Used Tableau to visualize findings and deliver data-driven recommendations.

Consulting Data Scientist (Project Partnership) Jan 2024 – May 2024
Lloyds Banking Group · Bristol, UK *University of Bristol Project*

- Analyzed transactional data to derive insights on customer behavior and fraud detection.
- Built a **Random Forest Classifier** improving classification of "Health & Wellness" and "Fashion & Apparel."
- Implemented **Isolation Forest** and **One-Class SVM** for fraud detection.
- Performed CLV calculations and segmentation for targeted marketing strategies.

Market and Competitor Analyst Intern Jan 2024 – Mar 2024
Albotherm · Bristol, UK *Internship*

- Conducted comprehensive market research and developed segmentation strategies.
- Synthesized competitive analysis into actionable insights for management.
- Presented findings using data visualization techniques for strategic decision-making.

Artificial Intelligence Consultant Nov 2022 – Jun 2023
DataMites · Bengaluru, India *Consulting*

- Built predictive models for healthcare and IoT datasets.
- Applied **GANs** for synthetic data generation and reinforcement learning for process automation.
- Performed fairness and interpretability analysis to ensure responsible AI deployment.

Student Intern Nov 2021 – Jan 2022
1Stop.ai · Remote *Internship*

- Developed CNN models for handwritten digit classification and celebrity image recognition.
- Applied NLP techniques for automated news categorization.
- Improved model performance via preprocessing, augmentation, and hyperparameter tuning.

Data Science Consultant Oct 2021 – May 2022
DataMites · Bengaluru, India *Consulting*

- Built a flight fare prediction model using **XGBRegressor**.
- Conducted EDA using SQL and Tableau for actionable insights.
- Supervised data analysts and acted as secondary account manager for clients.

Data Science Intern Apr 2021 – Jul 2021
Innomatics Research Labs · Remote *Internship*

- Built a Music Recommendation System using machine learning techniques.
- Deployed solutions using Flask for practical applications.
- Conducted EDA and visualization with Python for insights.

Projects

Real-Time Speech Intelligence & Voice Assistant | *Python*, *AssemblyAI*, *OpenAI API*, *Streamlit* 2025

- Built an end-to-end speech processing system supporting **audio upload, YouTube ingestion, and real-time microphone input**.
- Implemented a **chunk-based real-time transcription pipeline** using AssemblyAI, improving output quality via duration tuning and language constraints.
- Developed a **speech-to-insight pipeline** generating transcripts, sentiment labels, and structured summaries from long-form audio.
- Created an interactive **Streamlit web application** enabling users to analyze audio content with a single input interface.
- Integrated a **voice assistant using LLM APIs** to generate contextual responses from live speech input.
- Engineered file handling and session management to ensure reliable multi-input processing across workflows.

Manaswel – AI-Powered Mental Health Platform | *Python*, *PyTorch*, *HuggingFace*, *OpenCV*, *Flask*, *Edge Deployment* Feb 2025 – Jul 2

- Designed AI-driven interaction models combining NLP-based chat analysis with lightweight CV for behavioral monitoring.
- Explored **BERT**, **DistilBERT**, and **RAG** for context-aware, ethical response generation.
- Integrated feedback loops for emotional tone detection via user studies and accessibility research.
- Prototyped scalable edge-based architectures enabling real-time inference while ensuring privacy-preserving AI.
- Collaborated with mental health professionals and academic mentors to ensure responsible AI deployment.

Sentiment Analysis Using Transformers | *PyTorch*, *HuggingFace Transformers*, *AWS*

- Extended Transformer encoders to capture nuanced contextual relationships in text, including sarcasm and irony.
- Implemented an ML pipeline covering preprocessing, model training, hyperparameter tuning, dropout, and learning rate scheduling.
- Used AWS for dataset handling and scalable training, ensuring reproducibility and cloud integration.

Advanced Sentiment and Emotion Analysis in Tweets | *Transformers*, *BERT*, *DistilBERT*, *Attention Mechanisms*

- Developed an emotion classification system for Twitter data targeting emotions such as anger, joy, sadness, and optimism.
- Applied transfer learning and incorporated attention mechanisms for improved context understanding.
- Optimized training with early stopping, learning rate scheduling, and fine-tuning to avoid overfitting.
- Achieved strong performance on the TweetEval dataset, improving detection of ambiguous emotions.

AI-MedVision: AI for Medical Image Analysis | *TensorFlow*, *Keras*, *PyTorch*, *Streamlit*, *CNN*, *OpenCV*

- Developed CNN-based diagnostic models for COVID-19 detection (X-rays), brain tumor identification (MRI), and retinal disease classification.
- Applied data augmentation (rotation, scaling, flipping) to increase dataset diversity and reduce model bias.
- Deployed models with Streamlit, providing a real-time diagnostic support interface for healthcare professionals.
- Conducted extensive validation and testing on real-world datasets to evaluate reliability.

Vision Assist | *Python*, *YOLO*, *OpenCV*, *Speech Systems*, *Real-Time AI* Jan 2026 – Present

- Developing an accessibility-focused mobility assistance system using computer vision and AI for blind and visually impaired users.
- Building real-time object and hazard detection workflows with low-latency voice guidance using smartphone-based hardware.
- Exploring lightweight inference pipelines, speech-based alert systems, and human-centred AI interaction design for outdoor navigation support.
- Conducting accessibility-focused user research and mobility workflow analysis to improve usability and reduce cognitive load.

Conversational AI for PDFs and Websites | *RAG*, *LangChain*, *LlamaIndex*, *TruLens*, *OpenAI API*, *Flask/Streamlit*

- Developed Retrieval-Augmented Generation (RAG) pipelines to transform static documents and websites into conversational interfaces.
- Integrated **LangChain** and **LlamaIndex** for seamless indexing and querying of structured/unstructured data.
- Incorporated **TruLens** for interpretability, monitoring, and fine-tuning of model responses.
- Designed AI-driven workflows that improved accessibility of research papers, reports, and websites.

Courses and Certifications

GEN AI Mastermind - Growth school
AI Skills Challenge: Azure Machine Learning - Microsoft Build
Certified Data Scientist - DataMites
Design Thinking- NPTEL
Python for Data Science - NPTEL
Google Data Analytics - Coursera
Python 101 for Data Science- IBM

Education

University of Bristol Sep 2023 — Jan 2025
Master of Science in Data Science *Bristol, UK*

- **Relevant Coursework:** Software Development, Programming and Algorithms, Large-Scale Data Engineering, Visual Analytics, AI and Text Analytics, Ethics
- **Activities:** Google Developer Student Club, Data Science Society, Volunteer Coordinator for Women in Leadership
- **Achievements:** Bristol Plus award for exceptional performance, Top 100 in Google Solution Challenge and Hackathons winner.

Central University of Karnataka Jul 2018 — Aug 2022
Bachelor of Technology in Electronics and Communication Engineering (GPA: 3.31 / 4.0) *Kalaburagi, India*

- **Relevant Coursework:** Data Structures, Algorithms, Machine Learning, Signal Processing, Internet of Things
- **Activities:** Participated in academic and non-academic events, IEEE student member
- **Achievements:** Achieved first class grade

Competitions and Hackathons

IBM Call for Code - IBM
AI- Healthcare Hackathon - Encode
Google Developer Student Clubs (GDSC) 2024 - Google
Datathon - University of Bristol
BrisHack - University of Bristol
RSA Student Design Awards - University of Bristol

Activities

National Level Sports - England Kabaddi
Campus Ambassador - E-Summit, IIT Bhubaneswar
Bristol Student Volunteer - University of Bristol
Blogs Writting - Medium/Sudhamsr
Student Member - IEEE